PCA of ToF-SIMS spectra

Preprocessing transformations, All-peak PCA and Selected-peak PCA

November 5, 2026

Outline

1. **What PCA is**: Base definition
2. **The transformations** applied to the spectra before PCA: Bins, total-ion normalisation, square root, centring
3. **Two rounds of PCA** (Yang et al.): All-peak PCA, then Selected-peak PCA
4. **Our implementation** what differs from Yang

The problem: too many variables per sample

- A ToF-SIMS spectrum has hundreds of peaks. Each peak is one **variable**.
- A set example: **24 subsamples** $\times$ **901 bins** (1 u bins up to m/z 900).
- Question: which samples are chemically similar, and which peaks make them different?
- Looking at 24 spectra by eye does not answer this. A table of 24×901 numbers does not either.
- PCA is a standard way to summarise such data and is one of the most used methods for ToF-SIMS interpretation (Sek et al. 2021).

Data matrix X

$$X = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix}$$

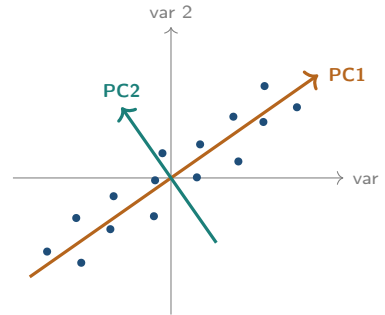
rows i: subsamples ($n = 24$)
columns j: peaks or bins ($p = 901$)
entry x_{ij}: ion counts of bin j in subsample i

PCA

Definition

PCA finds new axes, the **principal components** (PCs), that are linear combinations of the original variables.

- **PC1** is the direction with the largest variance in the data.
- **PC2** is the largest remaining direction, at a right angle to PC1. And so on.
- Each sample is projected on the first PCs. The result is a map where similar samples lie close together.
- It is **unsupervised**: Group labels are not used to compute the axes.



schematic: PC1 follows the long axis of the cloud

PCA

1. **Centre** each column: $Z = X - \mathbf{1}\,\bar{x}^\top, \quad \bar{x}_j = \frac{1}{n}\sum_i x_{ij}$
2. **Covariance** of the variables: $C = Z^\top Z/(n-1)$
3. **Directions**: eigenvectors $C\,v_k = \lambda_k v_k$, orthonormal, with $\lambda_1 \geq \lambda_2 \geq \ldots$
   Computed in practice by the singular value decomposition $Z = USV^\top$, with $\lambda_k = s_k^2/(n-1)$
4. **Scores**: coordinates of the samples, $T = ZV = US$ — one row per sample
5. **Explained variance**: $\mathrm{EV}_k = \lambda_k \big/ \sum_l \lambda_l$

Consequences

The loadings are the columns of V (one weight per peak and PC). At most $\min(n-1,\, p)$ components exist. Because PCA maximises variance, **variables with large variance dominate**, so the scaling of the data matters.

How to read a PCA result

Scores

One point per **subsample**, at its position on two PCs.

Points close together have similar spectra. Labels only colour the points.

Loadings

One weight per **peak** and PC.

A large $|\text{weight}|$ means the peak drives that axis. The sign gives the direction.

Explained variance

Share of the total variance carried by each PC (scree plot).

Tells how many PCs are worth looking at.

- The axes are **not** m/z. The m/z of a peak only appears in the loadings, where it says which peaks build each axis.
- Scores answer "which samples differ", loadings answer "which peaks make them differ".

Why spectra need preprocessing before PCA

PCA follows variance, so anything that inflates variance for the wrong reason distorts the result.

Different total signal

Total ions per subsample range from **18.8 M to 86.4 M** (4.6×) in our set. PC1 could just reflect signal size.

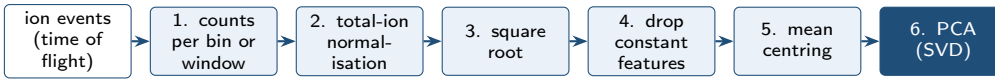
Range

A few intense low-mass peaks carry most of the counts and would dominate the variance.

Counting noise

Ion counts are Poisson-like: the larger the count, the larger its random spread.

The transformation pipeline (as implemented)



- Order is fixed: **normalise** → **square root** → **centre** (Yang et al. 2019).
- Steps 2 and 3 can be switched off in the app (`normalize = none`, `sqrt` off). Centring and the SVD always run.
- Peak-quality metrics (slide 19) are computed on the *raw* counts, before step 2.
- The same pipeline is used for the All-peak and the Selected-peak PCA. Only the features of step 1 differ.

Step 1: from ion events to a feature matrix

Each ion has a time of flight t; the calibrated mass is $m = \big((t - k_0)/s_f\big)^2$.

All-peak PCA

Every bin of width w is a feature:

$$x_{ij} = \#\{\text{ions of } i : \; j\,w \le m < (j+1)\,w\}$$

$w = 1$ u up to m/z 900 gives $p = 901$ bins. No peak identification, no selection.

Selected-peak PCA

Only chosen peak windows are features:

$$x_{ip} = \#\{\text{ions of } i : \; lo_p \le m < hi_p\}$$

one integrated count per peak, read directly from the time-of-flight channels.

Result: a matrix of raw counts X with one row per subsample, kept for export before any transformation.

Step 2: total-ion normalisation

Formula

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sum_k x_{ik}} \qquad \Rightarrow \qquad \sum_j \tilde{x}_{ij} = 1 \quad \text{for every subsample } i$$

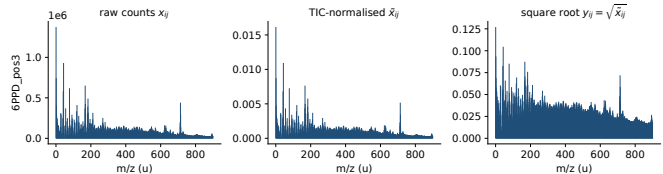
- Each spectrum is divided by its own total, so rows describe **composition**, not signal size.
- In the All-peak PCA the sum runs over all bins. In the Selected-peak PCA it runs over the **selected peaks only**, as in Yang et al. ("normalised to the selected total ion intensity").
- Our subsamples differ by up to 4.6× in total ions, which this step removes.
- App setting: `normalize = total-ion` (default) or `none`.

Step 3: square-root transform

Formula

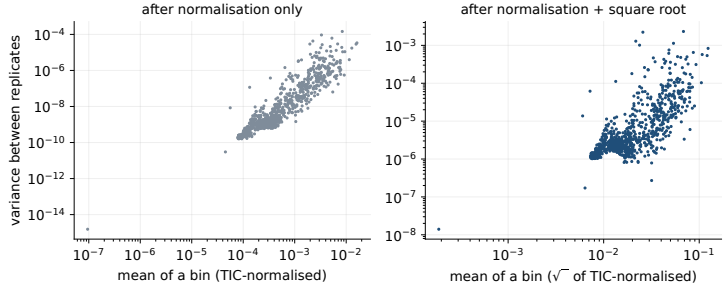
$$y_{ij} = \sqrt{\tilde{x}_{ij}}$$

- Compresses the **dynamic range**: big peaks shrink most, small peaks gain relative weight.
- **Variance stabilisation**: for Poisson counts the variance grows with the mean, while the variance of $\sqrt{\cdot}$ is roughly constant.
- App setting: `sqrt` on (default) or off.



One real spectrum (6PPD_pos1, 1 u bins) at three stages. The shape is preserved, the dominance of the largest bins is reduced.

Does the square root help on our data?



- Each dot is one of the 901 bins: mean and variance over replicates, averaged over the five sample groups.
- The spread of the variance (1st to 99th percentile) is **5.1 decades** after normalisation alone and **2.8 decades** after the square root.
- The variance is compressed but not made constant: replicates also differ for reasons beyond counting noise.

Steps 4 and 5: drop constant features, centre the columns

Drop constant features

$$\text{keep } j \iff s_j = \mathrm{sd}_i(y_{ij}) > 0$$

Bins or peaks with no variation carry no information and are removed. All 901 bins were kept in our run.

Mean centring

$$z_{ij} = y_{ij} - \bar{y}_j, \qquad \bar{y}_j = \tfrac{1}{n} \sum_i y_{ij}$$

Puts the origin at the average spectrum. PCs then describe **differences between samples**.

- **No unit-variance scaling** (z-score). Yang's recipe only centres. Dividing by s_j would give every peak equal weight and inflate weak, noisy peaks.
- The PCA itself is the SVD of Z (slide on PCA in five lines). Backend: GPU (cuML) if available, else scikit-learn, else numpy.

Worked example: 3 subsamples × 4 peaks

Raw counts x_{ij}

	P1	P2	P3	P4	total
S1	400	100	400	100	1000
S2	900	200	600	300	2000
S3	100	400	300	200	1000

Normalised $\tilde{x}_{ij}$

	P1	P2	P3	P4
S1	0.40	0.10	0.40	0.10
S2	0.45	0.10	0.30	0.15
S3	0.10	0.40	0.30	0.20

Square root y_{ij}

	P1	P2	P3	P4
S1	0.632	0.316	0.632	0.316
S2	0.671	0.316	0.548	0.387
S3	0.316	0.632	0.548	0.447

Centred z_{ij}

	P1	P2	P3	P4
S1	0.093	−0.105	0.056	−0.067
S2	0.131	−0.105	−0.028	0.004
S3	−0.224	0.211	−0.028	0.064

- S2 has twice the counts of S1, but after normalisation the two are close. S3 is different.
- PC1 explains **95.6%** of the variance and separates S3 from S1 and S2 (scores 0.155, 0.160, −0.315). With $n = 3$ at most two PCs exist.

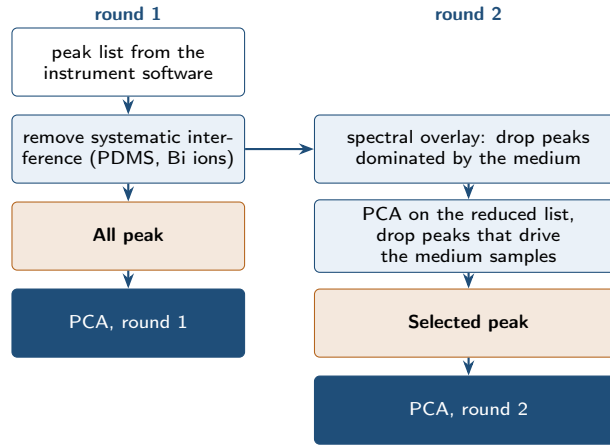
Summary of the transformations

Step	Formula	Purpose	App setting
Features	counts per bin or window	turn a spectrum into a row	bin width, peaks
Total-ion normalisation	$\tilde{x}_{ij} = x_{ij} / \sum_k x_{ik}$	removes total-signal differences	`normalize`
Square root	$y_{ij} = \sqrt{\tilde{x}_{ij}}$	tame dynamic range and counting noise	`sqrt`
Drop constant	keep j if $\mathrm{sd}(y_{\cdot j}) > 0$	remove empty features	automatic
Mean centring	$z_{ij} = y_{ij} - \bar{y}_j$	origin at the mean spectrum	always
PCA (SVD)	$Z = USV^\top$	axes of maximal variance	components

Not applied: logarithm, unit-variance scaling (z-score) and Pareto scaling. Only the steps above exist in the code.

Two rounds of PCA: the idea of Yang et al. (2019)

- Title: *Peak selection matters in principal component analysis*.
- Question: how much does the choice of peaks change a spectral PCA?
- Samples: two *Geobacter* species, alone and in coculture, plus two culture media (liquid ToF-SIMS).
- Same preprocessing in both rounds: normalise → square root → mean-centre.
- Round 1 uses **all peak**, round 2 uses **selected peak**.



What Yang removes, and how

Round 1: "all peak"

Known systematic interference is removed from the peak list:

- PDMS from the device: m/z$^+$ **147, 221, 281, 295, 369**
- Bi primary ions: m/z$^+$ **209, 418, 657**

Everything else is "all peak". Mass calibration comes first (positive mode: Si^+, Bi^+, Bi_2^+, Bi_3^+).

Round 2: "selected peak" (spectral overlay)

1. For every unit mass take the **maximum** normalised intensity over all samples.
2. If the **medium** sample reaches more than **90%** of that maximum and dominates the medium spectra, remove the peak.
3. Run PCA on the reduced list. Peaks that drive the medium samples in scores and loadings are removed too.
4. Keep characteristic peaks: amino acids, lipids, water clusters, fatty acids.

Yang et al.: what changed between the two rounds

All peak

- PC1 + PC2 explain **90.06%**.
- The two *Geobacter* populations separate.
- Medium salts dominate the high scores (K^+ m/z 39, $CaOH^+$ 57, Fe^+ 56) and hide the biology.

Selected peak

- PC1 + PC2 explain **72.67%**.
- The same populations separate.
- Microbial peaks (amino acids, lipids, DNA bases, water clusters) get higher loadings and the result is easier to interpret.

- Message: separation can look similar, but **which peaks explain it** changes with peak selection. Explained variance alone is not the goal.
- Sub-set PCAs (selected peaks, amino-acid peaks only, water-cluster peaks only) were all useful. High-intensity and high-resolution data both worked.

Selection criteria for the second round (Fu et al. 2018)

Same two-round scheme, with **rules** for the second PCA. Round 2: peaks are chosen if

(a) the loading of the peak is clearly higher than that of its neighbours,

(b) the peak is **at least 3×** more intense than its neighbours,

(c) the peak carries **more than 0.5%** of the total ion intensity,

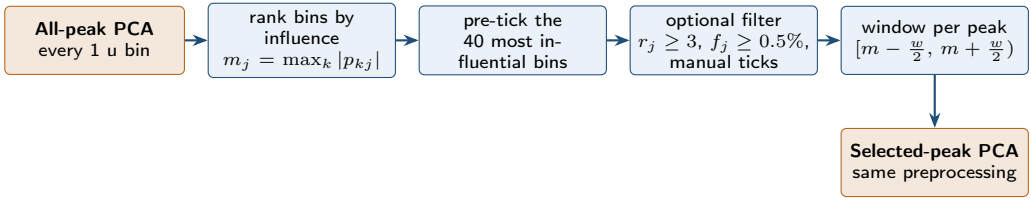
(d) product peaks identified in earlier papers are included.

The same tests in our tool (computed on raw counts)

$$r_j = \frac{I_j}{\max(I_{j-1}, I_{j+1})}, \qquad f_j = \frac{I_j}{\sum_i N_i}, \qquad \text{pass} \iff r_j \geq 3 \text{ and } f_j \geq 0.5\%$$

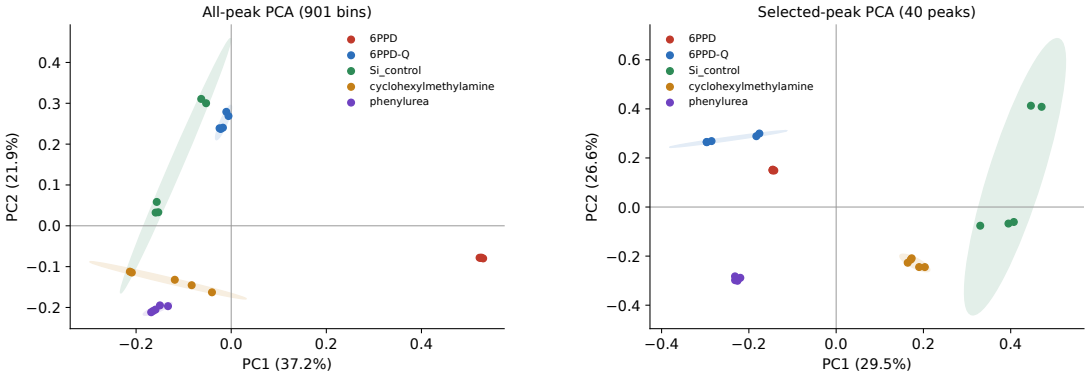
I_j: ions of bin j over all subsamples, N_i: total ions of subsample i. Defaults can be changed per run.

Our implementation: All-peak PCA, then Selected-peak PCA



- The user can add or remove peaks and subsamples by hand. The choice is passed live to the second tab.
- Both PCAs give scree, scores of every PC pair with 90% ellipses per label group, and loadings.

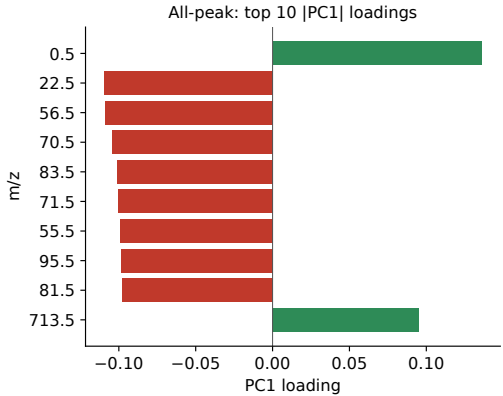
Real data: scores, All-peak vs Selected-peak

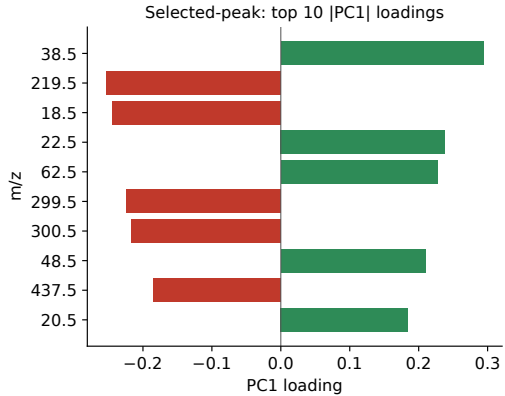


24 subsamples, positive polarity, normalised, square root, mean-centred. Shaded ellipse: 90% of each label group (groups with fewer than three points or on a line have none).

- The five groups are distinguishable in both PCAs. In the All-peak PCA, 6PPD is far from the rest on PC1.

Real data: loadings of PC1

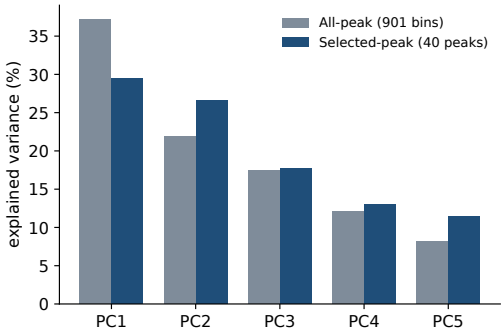




m/z is the bin centre. Green: positive loading, red: negative.

- Loadings tell which bins separate the samples on PC1. Their identity (which ion) is then checked against the peak table.

Real data: numbers



	All-peak	Selected-peak
features	901 bins	40 peaks
PC1	37.2%	29.5%
PC2	21.9%	26.6%
PC1 + PC2	59.1%	56.1%
PC1 to PC5	96.8%	98.2%

The 40 selected bins hold 17.6% of all ions.

How our tool differs from Yang et al.

	Yang et al. 2019	**This tool**
Mass calibration	done in the instrument software	done upstream, data already calibrated
Interference (PDMS, Bi)	removed before "all peak"	**not removed**: every bin is a feature
Medium overlay ($> 90\%$)	used to build "selected peak"	**not implemented** (needs a medium or matrix sample)

- The selected peaks are picked from the same data they are then analysed on.

Take-home

- **PCA** turns a spectrum table into a few axes: **scores** show which samples differ, **loadings** show which peaks make the difference.
- Spectra are first transformed: **total-ion normalisation** $\rightarrow$ **square root** $\rightarrow$ **mean centring**. No z-score, no logarithm.
- **Two rounds** (Yang et al.): All-peak PCA first, then Selected-peak PCA on a reduced, curated peak list. **Peak selection changes the interpretation**.
- Selection tests (Fu et al.): $\geq 3\times$ neighbours and $> 0.5\%$ of the ions, implemented as r_j and f_j.

References

- C. Yang, W. Wei, F. Liu, X.-Y. Yu. Peak selection matters in principal component analysis: A case study of syntrophic microbes. *Biointerphases* 14, 051004 (2019). doi:10.1116/1.5118237

- Y. Fu, Y. Zhang, F. Zhang, J. Chen, Z. Zhu, X.-Y. Yu. Does interfacial photochemistry play a role in the photolysis of pyruvic acid in water? *Atmospheric Environment* 191, 36–45 (2018). doi:10.1016/j.atmosenv.2018.07.061

- K. L. Sek, P. L. Lee, K. Y. Pang, Y. Hua, L. Zhu, X. Li. Principal Component Analysis (PCA) of Surface Contamination by TOF-SIMS. *IEEE IPFA* (2021). doi:10.1109/IPFA53173.2021.9617378